



Servlet I

1. Servlet basics
2. Creating simple servlet
3. Web application structure
4. Deployment and execution
5. Servlet life cycle
6. Servlet API

- What is servlet
 - It is java class used for generating dynamic HTML
 - It gets executed after receiving the request from client
 - Life cycle of servlet is under the control of web server where it is hosted
- Web server
 - A web server is software that allows hosting of web applications.
 - Major task of web server is the client communication i.e accepting request and sending response
- Static web page
 - When a site visitor requests a static page, say, by clicking a link, selecting a browser bookmark, or entering a URL; the web server sends the page directly to the web browser.
 - Response is same for all users

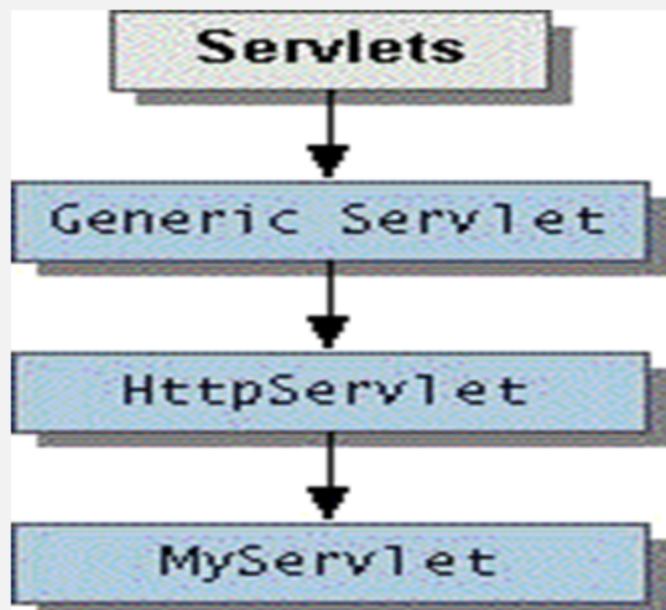
- Dynamic web page
 - A dynamic page displays different content for different users with the same layout and design.
 - e.g. weather updates or stock prices
- Web container
 - Web Container is a part of Web Server
 - It mainly maps incoming requests to correct servlet and manages servlet's life cycle
- Examples of open source web containers:
 - Apache tomcat
 - Glassfish
 - JBoss.

- Create dynamic web project in eclipse IDE.
- In the src folder add the servlet class in the appropriate package
 - It should implement Servlet interface directly or indirectly
 - Override doGet() or doPost() method to generate response
- @WebServlet annotation:
 - Servlet 3.0 API has provided with different annotations.
 - @WebServlet annotation is written baove servlet class for mapping request url.
 - Same configuration can be done in web.xml deployment descriptor

- Root directory of web application consists of mainly WEB-INF directory
- WEB-INF directory is composed of
 - classes directory - consists of compiled servlet classes in the appropriate package structure
 - lib directory - third party jar files
 - web.xml deployment descriptor - XML based file
- HTML/JSP pages can reside directly in the root folder or can be added in separate folders within the root directory

- **Step 1 : Download and unzip tomcat server**
- After unzipping server folder should have lib, bin and webapps directory mainly with some additional folders
- **Step 2 : Linking apache tomcat to eclipse environment**
- Add tomcat runtime environment from windows-> preferences menu
- **Step 3 : Creating local instance of tomcat server in eclipse**
Select the deploy path of server as tomcat's deployment directory
- **Step 4: Testing the servlet**
Start the server
Open the browser and hit the URL
- **Step 5: Verifying deployed application** in tomcat deployment directory

- The life cycle of a servlet can be summarized as:
 1. The Servlet class is loaded.
 2. The Servlet instance is created
 3. The `init()` method is invoked in order to initialize the servlet.
 4. The `service()` method is invoked repeatedly for each client request placed.
 5. The servlet is destroyed using `destroy()` method.



- The Servlet interface is the root interface of the servlet class hierarchy.
- User defined servlets can extend from GenericServlet or HttpServlet.
- doGet() or doPost() method receive 2 arguments as request and response objects

- HttpServletRequest extends ServletRequest interface and adds few HTTP related methods
- Similarly HttpServletResponse extends ServletResponse interface and adds few HTTP related methods

